

Demarcus Williams Homework 1.

$$(1) \therefore R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

$$R_x(-\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(-\theta) & -\sin(-\theta) \\ 0 & \sin(-\theta) & \cos(-\theta) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{bmatrix}$$

$$R_x^{-1}(\theta) = R_x^T(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{bmatrix}$$

$$\therefore R_x^{-1}(\theta) = R_x(-\theta)$$

$$\therefore R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \end{bmatrix}$$

$$\text{iv. } R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$R_y(-\theta) = \begin{bmatrix} \cos(-\theta) & 0 & \sin(-\theta) \\ 0 & 1 & 0 \\ -\sin(-\theta) & 0 & \cos(-\theta) \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$R_y^{-1}(\theta) = R_y^T(\theta) = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$\therefore R_y^{-1}(\theta) = R_y(-\theta)$$

$$\therefore R^{-1}(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \end{bmatrix}$$

$$\begin{bmatrix} \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_z(-\theta) = \begin{bmatrix} \cos(-\theta) & -\sin(-\theta) & 0 \\ \sin(-\theta) & \cos(-\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_z^{-1}(\theta) = R_z^T(\theta) = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore R_z^{-1}(\theta) = R_z(-\theta)$$

$$\therefore R_m(-\theta) = R_m^{-1}(\theta) \quad m = x, y, z$$

$$\textcircled{2}. R_y = \begin{bmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{bmatrix}$$

$$R_z = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R = R_y(\phi) R_z(\theta)$$

$$= \begin{bmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos \phi \cos \theta & -\cos \phi \sin \theta & \sin \phi \\ \sin \phi \cos \theta & -\sin \phi \sin \theta & \cos \phi \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos \phi \cos \theta & \sin \phi \cos \theta & \sin \theta \\ \sin \theta & \cos \theta & 0 \\ -\sin \phi \cos \theta & \cos \phi \sin \theta & \cos \theta \end{bmatrix}$$

$$R' = R_z(\theta) R_y(\phi)$$

$$= \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{bmatrix}$$

$$= \begin{bmatrix} \cos \phi \cos \theta & -\sin \theta & \sin \phi \cos \theta \\ \cos \phi \sin \theta & \cos \theta & \sin \phi \sin \theta \\ -\sin \phi & 0 & \cos \phi \end{bmatrix}.$$

$$R'^T(-\theta, -\phi) = \begin{bmatrix} \cos(-\phi) \cos(-\theta) & -\sin(-\theta) & \sin(-\phi) \cos(-\theta) \\ \cos(-\phi) \sin(-\theta) & \cos(-\theta) & \sin(-\phi) \sin(-\theta) \\ -\sin(-\phi) & 0 & \cos(-\phi) \end{bmatrix}^T$$

$$= \begin{bmatrix} \cos \phi \cos \theta & \sin \theta & -\sin \phi \cos \theta \\ -\cos \phi \sin \theta & \cos \theta & \sin \phi \cos \theta \\ \sin \phi & 0 & \cos \phi \end{bmatrix}^T$$

$$= \begin{bmatrix} \cos \phi \cos \theta & -\cos \theta \sin \theta & \sin \phi \\ \sin \theta & \cos \theta & 0 \\ \sin \phi \cos \theta & \sin \phi \cos \theta & \cos \phi \end{bmatrix}$$

$$= R$$

$$\therefore R(\theta, \phi) = R'(-\theta, -\phi)^T$$

(3). Roll - Pitch - Yaw

$$R(\phi) = R_z(\varphi) R_y(\theta) R_x(\psi)$$

$$= \begin{bmatrix} C_\varphi C_\theta & C_\varphi S_\theta S_\psi - S_\varphi C_\psi & C_\varphi S_\theta C_\psi + S_\varphi S_\psi \\ S_\varphi C_\theta & S_\varphi S_\theta S_\psi + C_\varphi C_\psi & S_\varphi S_\theta C_\psi - C_\varphi S_\psi \\ -S_\theta & C_\theta S_\psi & C_\theta C_\psi \end{bmatrix}$$

$$\begin{bmatrix} -\sin\theta & \cos\theta & \cos\psi \\ \cos\theta & \sin\theta & -\sin\psi \\ \sin\theta & \sin\theta & \sin\psi \end{bmatrix}$$

$$L_\theta = 0 \Rightarrow \left\{ \frac{\pi}{2}, \frac{3\pi}{2} \right\}$$

$$\text{case } \theta = \frac{\pi}{2}$$

$$= \begin{bmatrix} 0 & \cos\psi \sin\psi - \sin\psi \cos\psi & \cos\psi \cos\psi + \sin\psi \sin\psi \\ 0 & \sin\psi \sin\psi + \cos\psi \cos\psi & \sin\psi \cos\psi - \cos\psi \sin\psi \\ -1 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & \sin(\psi - \psi) & \cos(\psi - \psi) \\ 0 & \cos(\psi - \psi) & \sin(\psi - \psi) \\ -1 & 0 & 0 \end{bmatrix}$$

$$\alpha = \psi - \psi$$

case $\Theta = \frac{3\pi}{2}$

$$= \begin{bmatrix} 0 & -C\varphi S\varphi - S\varphi C\varphi & -C\varphi C\varphi + S\varphi S\varphi \\ 0 & -S\varphi S\varphi + C\varphi C\varphi & -S\varphi C\varphi - C\varphi S\varphi \\ +1 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -\sin(\varphi + \varphi) & -\cos(\varphi + \varphi) \\ 0 & \cos(\varphi + \varphi) & -\sin(\varphi + \varphi) \\ -1 & 0 & 0 \end{bmatrix}$$

$$\alpha = \varphi + \varphi$$

$\therefore$ for $\cos \alpha = 0$ $\alpha \approx \varphi \pm \varphi$

φ & φ are coupled & cannot be uniquely determined so there are multiple pairs that could work.

One angle could be chosen freely & then the final one may be determined.